

RUBRICEYE EXAM

INTEGRATED SCIENCE & COMPUTING — MOCK EXAMINATION

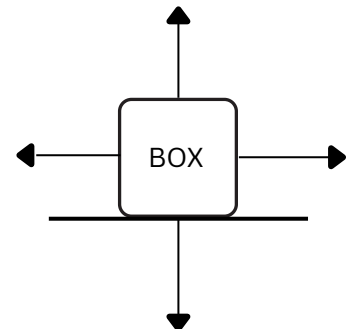
Candidate Name: _____ Candidate ID: _____ Date: _____
Time: 60 Minutes Maximum Marks: 35

SECTION B – Short Answer Questions (20 Marks)

Q2: Write your answers in the spaces provided. Attempt Only 5 Questions.

- State Newton's Second Law of Motion and write its mathematical expression. [4]
- Differentiate between RAM and ROM. Give one example of each. [4]
- Explain the role of the nucleus in a cell. [4]
- What is the difference between a series and a parallel circuit? Also Draw a diagram. [4]
- Name two input devices and two output devices used in a computer system. And what operation do they perform. [4]
- A force acts on a box on a horizontal surface. Study the given force diagram and identify the forces acting upward, downward, forward and backward. Complete the diagram with proper labeling. [4]
- Draw a bar graph representing the following data: [4]

Student	Marks (Total: 50)
Ahmed	45
Sara	20
Ali	35
Khadija	40



SECTION C – Structured / Extended Response (15 Marks)

Answer any 1 of the questions. Write in full sentences where required.

Q3: Attempt Both Parts of the following Question: [8+7]

- Explain the laws of reflection of light. In your answer, define the incident ray, reflected ray, normal, angle of incidence, and angle of reflection. State the relationship between the angle of incidence and the angle of reflection.
- Draw a complete ray diagram showing the reflection of a light ray from a plane mirror. Your diagram must clearly show and label the plane mirror, incident ray, reflected ray, normal, angle of incidence, and angle of reflection.

Q4: Attempt Both Parts of the following Question: [8+7]

- What is an algorithm? Explain any three characteristics of a good algorithm. Also state one advantage of writing an algorithm before solving a programming problem.
- Draw a complete flowchart to find and display the largest of three numbers entered by the user. Use the correct standard flowchart symbols, decision branches, and arrows.